

ELDCF: A Mathematical Consistency Framework for Physical AI

A fundamental vulnerability in modern robotics is that systems often slow down, fail, or shut down when they encounter complex, unstructured physical tasks: uneven ground, sudden loss of friction, stairs, sharp turns, or a carried object that begins to move.

The challenge is not simply object recognition. A humanoid robot must continuously judge whether its body, sensors, contact forces, motion plan, and environment can still be explained by one coherent physical situation.

Ludwig Wittgenstein wrote in *Tractatus Logico-Philosophicus* that the world is not merely a collection of things, but a totality of facts. This is a useful intuition for robotics. A robot does not act safely because it has identified isolated objects. It acts safely when it recognizes meaningful relations and events:

- a foot is in contact with the ground;
- the body is accelerating;
- the expected support force is present;
- the left and right sides of a gait remain coordinated;
- an IMU reading agrees with joint kinematics and contact measurements.

The engineering task is therefore to make complexity simpler without pretending that it has disappeared. We need compact models that preserve the relationships necessary for safe action.

Claude Shannon provided an important precedent. In his 1937 master’s thesis, later published in 1938 as *A Symbolic Analysis of Relay and Switching Circuits*, Shannon showed that complex relay networks could be represented and manipulated through Boolean algebra. Ten years later, information theory provided a mathematical language for uncertainty, communication, and the limits of reliable transmission.

Physical AI needs a comparable system-engineering language: one that can connect symbolic state, continuous mechanics, geometry, sensing, and coordination.

ELDCF is a proposed framework in that direction. It combines Boolean logic, Euler–Lagrange dynamics, Cartesian contact geometry, and graph signal processing into one interpretable safety-monitoring layer.

1 Mathematical structure

1.1 Boolean logic: compact contact and event states

Continuous measurements can be converted into logical states for contact, direction, and phase:

$$b_{\text{contact}} = \begin{cases} 1, & |F_z| > \varepsilon_F, \\ 0, & |F_z| \leq \varepsilon_F. \end{cases}$$

This does not discard the underlying force measurement. It provides a compact symbolic language for questions such as: *Is the foot supporting the robot? Has contact been lost? Is this motion phase physically valid?*

1.2 Euler–Lagrange dynamics: physical consistency

The robot’s measured motion should remain compatible with rigid-body dynamics:

$$r_{\text{EL}} = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) - \tau - J(q)^{\text{T}}F.$$

Here, q is robot configuration, $M(q)$ is the mass matrix, $C(q, \dot{q})\dot{q}$ represents velocity-dependent effects, $G(q)$ is gravity, τ is actuator torque, $J(q)$ is the contact Jacobian, and F is contact force.

The residual r_{EL} is not a controller. It is evidence of whether the current motion, force, and model can still be explained together.

1.3 Cartesian geometry: contact and friction

For a supporting foot, the relationship between tangential force, normal force, and friction matters:

$$\gamma = \frac{\|F_t\|}{\mu \max(F_n, F_{\min})}.$$

Here, F_t is tangential contact force, F_n is normal contact force, and μ is the assumed friction coefficient. A high value of γ indicates that the robot may be approaching friction saturation or slip.

1.4 Graph signal processing: structural coordination

A humanoid body can be represented as a graph whose nodes are meaningful subsystems and whose edges encode physically justified relationships: left–right correspondence, torso–leg coupling, or gait-phase coordination.

A symmetry-aware coordination residual may be written as:

$$r_{\text{sym}}(t) = \|x_{\text{left}}(t) - \mathcal{R} x_{\text{right}}(t - \Delta_{\phi})\|.$$

This can reveal when body segments no longer move in an expected coordinated relationship. Graph and frame-based features are optional diagnostic tools; they do not replace physics, contact checks, or state estimation.

2 ELDCF as a pre-trust layer

ELDCF combines these sources of evidence into a risk representation:

$$r(t) = \begin{bmatrix} r_{\text{EL}}(t) \\ r_{\text{contact}}(t) \\ r_{\text{observation}}(t) \\ r_{\text{symmetry}}(t) \\ r_{\text{recovery}}(t) \end{bmatrix}.$$

It can then provide supervisory outputs such as

{CONTINUE, SLOW, RECOVER_STEP, REPLAN, SAFE_STOP}.

ELDCE does not replace a world model, reinforcement learning, model predictive control, whole-body control, or low-level motor control. Those systems decide and execute actions.

ELDCE asks a prior question:

Before the robot commits to its next action, do its sensors, contact forces, body geometry, and physical dynamics still describe one coherent world state?

3 Research status

ELDCE is a research framework. Current evidence comes from synthetic and reduced-order tests of contact disturbances and observation faults. It does not yet guarantee that a humanoid will not fall or safely recover in every environment. Full-body simulation and hardware validation are required.